Tab 1

Target Audience for the Tool / Feature:

- Facilitators / Teacher who need more support in teaching to students

Current Project Detection

My current focus for this side of the project is on defining how a facilitator-supported educational system should operate in underserved and lower-resource learning environments, particularly within K–12 education. Rather than focusing purely on AI-driven teaching, the project aims to investigate how technology can support facilitators in delivering more structured, consistent, and accessible learning experiences.

A major issue identified through the literature review and stakeholder discussions is the shortage of qualified teachers and the resulting inconsistencies in educational quality, especially in rural or lower-income communities. In many cases, facilitators or parents may be required to teach subjects outside of their expertise, leading to inconsistent curriculum delivery and reduced confidence in lesson facilitation. Existing AI educational tools often assume that students are capable of self-directed learning; however, younger learners typically require significantly more structure, guidance, and support.

As a result, the system is being designed around structured lesson delivery and facilitator support rather than fully autonomous AI teaching. The educational purpose of the system is to guide facilitators through lessons, reduce uncertainty in delivery, support mixed-ability classrooms, and improve accessibility to quality educational experiences.

AI Integration Proposal

An important part of the project is determining the appropriate and ethical role of AI within educational environments. Rather than replacing facilitators or making autonomous educational decisions, the intention is for AI to act as a lightweight support system that assists facilitators in understanding and responding to classroom needs more effectively.

Current discussions have focused on the idea of an **AI-Assisted Facilitator Support and Learner Readiness System**. The goal of this prototype is to help facilitators quickly understand student support needs before and during a school term, particularly in classrooms with varying ability levels or limited educational resources.

- From teachers + outside perspectives

Instead of relying on rigid “learning style” classifications, the system would focus on identifying behavioural and engagement indicators that may help facilitators adapt lesson delivery more effectively.

Proposed Prototype Concept

The proposed prototype would involve students completing a short onboarding or learner-readiness assessment at the beginning of a term. This assessment would include a variety of simple educational activities such as reading tasks, visual exercises, interactive activities, pacing-based tasks, and short problem-solving exercises appropriate to different age groups.

During these activities, the system would collect interaction data including:

- completion time,
- retry attempts,
- hint usage,
- interaction frequency,
- task preference,
- and progression trends.

This data would then be analysed using lightweight AI or machine learning techniques to identify potential engagement patterns, pacing needs, support indicators, and instructional response trends.

The resulting outputs would not attempt to diagnose or categorise students, but instead provide facilitators with practical and interpretable support insights. For example, facilitators may receive recommendations such as:

- students requiring shorter instruction blocks,
- students benefiting from guided examples,
- or groups requiring additional pacing support.

These outputs would be displayed through a facilitator dashboard containing learner baseline summaries, pacing recommendations, engagement concerns, and suggested facilitation strategies.

Machine Learning Opportunities

From a technical perspective, this also creates opportunities to explore lightweight and explainable machine learning approaches. Potential inputs include behavioural metrics and activity interaction data collected during onboarding activities.

Possible approaches currently being explored include:

- clustering methods to identify groups of students with similar interaction patterns,
- simple classification models,
- or recommendation systems that generate facilitator support suggestions.

Importantly, the focus is on producing interpretable and facilitator-supportive outputs rather than opaque or fully autonomous AI systems.

Current Design Focus

The current design direction prioritises:

- structured facilitator-guided learning,
- accessibility in underserved educational environments,
- ethical AI integration,
- and lightweight learner support tools that reduce facilitator workload without replacing human oversight.

The intention is to ensure that AI remains a supportive educational tool rather than becoming the primary driver of instruction.

Immediate Next Steps

The current next steps for the project include:

1. Defining the onboarding assessment activities for different age groups.
2. Determining which learner interaction metrics are most useful and ethical to collect.
3. Designing the outputs and facilitator recommendations generated by the system.
4. Selecting appropriate machine learning approaches for the prototype.
5. Designing how facilitators interact with and utilise these insights within the platform.

A facilitator-support platform that uses lightweight AI-assisted learner interaction analysis to improve structured educational delivery in underserved K–12 environments.

Questions:

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Define how the educational system should work, why it should work that way, and how AI can support facilitators responsibly.

- educational logic
- facilitator workflows
- AI integration reasoning
- user needs
- research-backed design decisions

Current Issues:

- teacher shortages
- inconsistent educational delivery
- lack of facilitator support
- low-resource learning environments
- mixed ability classrooms

Current System Failures

- younger learners need structure
- facilitators lack confidence
- existing AI tools assume self-directed learning
- inconsistent curriculum delivery

System Support / Purpose (Educationally)

- guide facilitators
- provide structured lessons
- reduce uncertainty
- support mixed-ability classrooms
- use AI carefully

What should AI actually do?

- where AI helps
- where AI should NOT be used
- ethical limits
- human oversight

Goal / Prototype:

Build one strong, focused AI prototype that directly supports facilitators in underserved educational environments. → AI-Assisted Facilitator Support & Learner Readiness System

- Help facilitators quickly understand student support needs before and during a term.

Student completes:

A short onboarding assessment:

- reading activity
- visual task
- interactive task
- short quiz/problem solving

- pacing-based exercises

System collects:

- completion time
- retries
- hint usage
- interaction frequency
- task preference
- progression trends

AI/ML layer analyses:

- engagement patterns
- pacing needs
- support indicators
- instructional response patterns

Facilitator dashboard shows:

- learner baseline summary
- suggested facilitation strategies
- engagement concerns
- pacing recommendations

Machine Learning Opportunities:
Inputs:

- behavioural metrics
- activity interaction data

Model:

- clustering
- simple classifiers
- recommendation engine

Outputs:

- learner support profiles
- facilitator suggestions

Next Steps:
1. What onboarding activities exist?
2. What metrics are collected?
3. What outputs are generated?
4. What ML approach is used?
5. How do facilitators see/use the results?

Existing Technologies: From Inter Expo

- Linewise (Supported by Qoria) → Digital safety for children
- Saasyan (AI-Powered Student Safety for K-12 Schools)
- KaiLab (Teaching coding to students)

Tab 2

Research Areas:

- ❖ What decisions do facilitators actually make?
  - ➢ What information would help a less experienced facilitator teach more effectively?

Research:

- formative assessment
- teacher decision-making
- classroom intervention strategies
- mixed-ability classrooms
- instructional differentiation

Questions:

- How do teachers identify struggling students?
- How do they adapt lessons?
- What indicators do they look for?
- What interventions do they use?

- ❖ Learning Analytics
  - ➢ What data do we need to collect from students?

Search terms:

- Learning Analytics
- Educational Data Mining
- Student Engagement Analytics
- Classroom Analytics

Questions:

- What learner interaction data is commonly collected?
- What metrics are meaningful?
- What metrics are ethical?

Example metrics:

- completion time
- attempts
- retries
- hint usage
- task completion
- Progression

- ❖ Explainable AI in Education

Research:

- Explainable AI (XAI)
- Human-in-the-loop AI
- AI decision support systems
- Educational AI ethics

Questions:

- How do we explain recommendations?

- How do we avoid black-box decisions?
- How do facilitators remain responsible?

You want to be able to say:

> The system provides recommendations rather than decisions.

❖ Intervention Strategies

Research:

- instructional scaffolding
- worked examples
- mastery learning
- differentiated instruction
- cognitive load theory

❖ Baseline Assessment Design

Research:

- diagnostic assessment
- learner readiness assessment
- baseline testing

Questions:

- What activities should students complete?
- How long should onboarding take?
- What can realistically be measured?

I would aim for:

Reading Task

Measures:

- comprehension
- pace

Quiz Task

Measures:

- accuracy
- confidence

Guided Task

Measures:

- hint usage

Problem Solving Task

Measures:

- Persistence

❖ Facilitator Dashboard Design
  ➢ What information does the facilitator see?

Research:

- teacher dashboards
- learning analytics dashboards
- Moodle analytics
- Canvas analytics
- Khan Academy teacher dashboard

Questions:

What should a facilitator see in under 30 seconds?

Class Overview

- students needing support
- students progressing normally
- students needing challenge

Student Card

Strengths

Support Needs

Recent Trends

Recommended Actions

AI Assistant

Questions:

- Who needs support?
- What should I focus on today?
- Which students are disengaged?

Question 1

Should the project focus on identifying learner needs, recommending interventions, or both?

Question 2

Is the key contribution the analytics layer, the facilitator workflow, or the AI recommendation system?

Question 3

Would a rules-based recommendation engine be sufficient for the prototype, or should we investigate machine learning approaches?

Question 4

Should the onboarding assessment remain a one-off diagnostic activity, or should data be collected continuously throughout the term?

Question 5

What would constitute a successful facilitator-support recommendation from an educational perspective?

requirements elicitation

stakeholder analysis

system architecture

AI integration design

data pipeline design

dashboard design

explainable recommendation systems

human-computer interaction

evaluation of software usefulness

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Interfaces:

- Teacher
  - How to integrate with Moksh side
  - Software side (most classrooms and have computer labs + phones)
  - Focus on the design and concept
  - Deep into one area → for the scope of the project (demonstrate that you can go deeper into → good for research project)
    - Constraining the problem to be reasonable (quality over quantity)
    - Going deeper is better than spreading thin
- Student
  - Seperate
  - Research into what is good practice
  - Trying out ideas
- Solving two parts of the same problem

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